



CARIBBEAN EXAMINATIONS COUNCIL

ADVANCED PROFICIENCY EXAMINATION

PHYSICS

UNIT 2 – Paper 02

2 hours 30 minutes

READ THE FOLLOWING INSTRUCTIONS CAREFULLY

1. This paper consists of **SIX** questions.
2. Section A consists of **THREE** questions. Candidates must attempt **ALL** questions in this section. Answers for this section must be written in this answer booklet.
3. Section B consists of **THREE** questions. Candidates must attempt **ALL** questions in this section. Answers for this section must be written in the separate answer booklet provided.
4. All working **MUST** be **CLEARLY** shown.
5. The use of non-programmable calculators is permitted, but candidates should note that the use of an inappropriate number of figures in answers will be penalised.

NOTHING HAS BEEN OMITTED.

LIST OF PHYSICAL CONSTANTS

Speed of light in free space	c	=	$3.00 \times 10^8 \text{ m s}^{-1}$
Permeability of free space	μ_0	=	$4\pi \times 10^{-7} \text{ H m}^{-1}$
Permittivity of free space	ϵ_0	=	$8.85 \times 10^{-12} \text{ F m}^{-1}$
Elementary charge	e	=	$1.60 \times 10^{-19} \text{ C}$
The Planck's constant	h	=	$6.63 \times 10^{-34} \text{ J s}$
Unified atomic mass constant	u	=	$1.66 \times 10^{-27} \text{ kg}$
Rest mass of electron	m_e	=	$9.11 \times 10^{-31} \text{ kg}$
Rest mass of proton	m_p	=	$1.67 \times 10^{-27} \text{ kg}$
Acceleration due to gravity	g	=	9.81 m s^{-2}
1 Atmosphere	Atm	=	$1.00 \times 10^5 \text{ N m}^{-2}$
Avogadro's constant	N_A	=	$6.02 \times 10^{23} \text{ per mole}$

SECTION A

Answer ALL questions.

You MUST write your answers in this answer booklet.

- 1.** (a) (i) With the aid of a diagram explain the meaning of the term 'magnetic flux'.

[2 marks]

- (ii) How is the unit of magnetic flux density related to the tesla?

[1 mark]

- (iii) State Faraday's law of electromagnetic induction.

[1 mark]

GO ON TO THE NEXT PAGE

- (b) Figure 1 shows a coil with 16 turns which is wound on top of a solenoid with a cross-sectional area of 50 cm^2 . An a.c. current through the solenoid produces a peak magnetic field of 7.5 mT which is changing at a rate of 500 Hz .

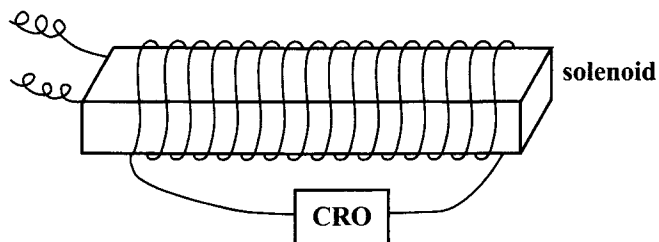


Figure 1

- (i) Calculate the MAXIMUM value of the TOTAL flux linkage through the 16 turn coil.

[3 marks]

- (ii) Find the peak value of the e.m.f. induced in the coil, given that the equation for the variation of the flux linkage is $\Phi = \Phi_0 \sin 2\pi ft$.

[2 marks]

- (iii) Figure 2 (a) shows the screen of the cathode ray oscilloscope connected to the coil.

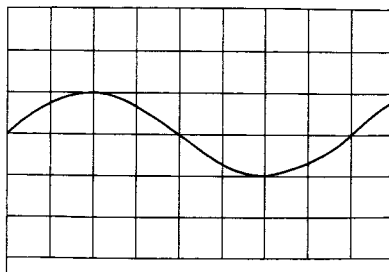


Figure 2 (a)

Show on Figure 2 (b) the c.r.o. trace you would expect if the number of turns wrapped around the solenoid was increased to 32.

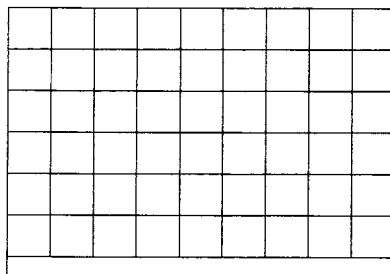


Figure 2 (b)

[2 marks]

On the third screen (Figure 2 (c)) show the effect of changing the frequency of the supply to the solenoid to 1000 Hz while **all other** factors remain constant.

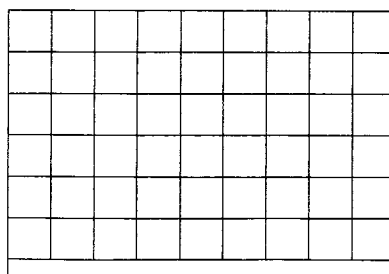


Figure 2 (c)

[2 marks]

- (iv) Describe briefly how you could show that the induced e.m.f depends on the area of the coil.

[2 marks]

Total 15 marks

2. (a) The circuit diagram in Figure 3, shows an operational amplifier connected as a non-inverting amplifier.

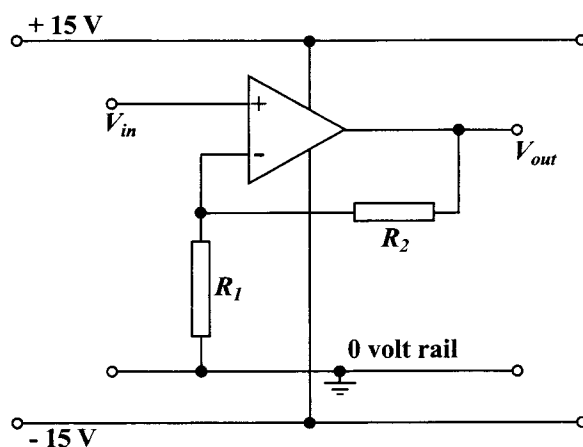


Figure 3. Operational amplifier connected as a non-inverting amplifier

- (i) Suggest a pair of values for the resistors, R_1 and R_2 , if the amplifier is to have a gain of +10.

$R_1 =$ _____ $\text{k}\Omega$ $R_2 =$ _____ $\text{k}\Omega$
[2 marks]

- (ii) What extra equipment would be needed to obtain data to plot the amplifier's transfer characteristic (graph of V_{out} vs V_{in})? Show on Figure 3 how these components would be connected.

[4 marks]

- (b) (i) Use the data in Table 1 to plot, on the grid on page 9, the transfer characteristic for a non-inverting amplifier similar to the one in part (a) on page 7.

TABLE 1

V_{in}/V	V_{out}/V	V_{in}/V	V_{out}/V
3.00	12.90	0.18	1.74
2.60	12.90	-0.59	-5.82
2.30	12.90	-0.98	-9.72
1.90	12.90	-1.18	-11.70
1.22	12.10	-1.60	-13.80
1.05	10.40	-1.90	-13.80
0.73	7.27	-2.60	-13.80
0.56	5.50	-3.00	-13.80

[5 marks]

- (ii) From your graph determine the gain of the amplifier.

[2 marks]

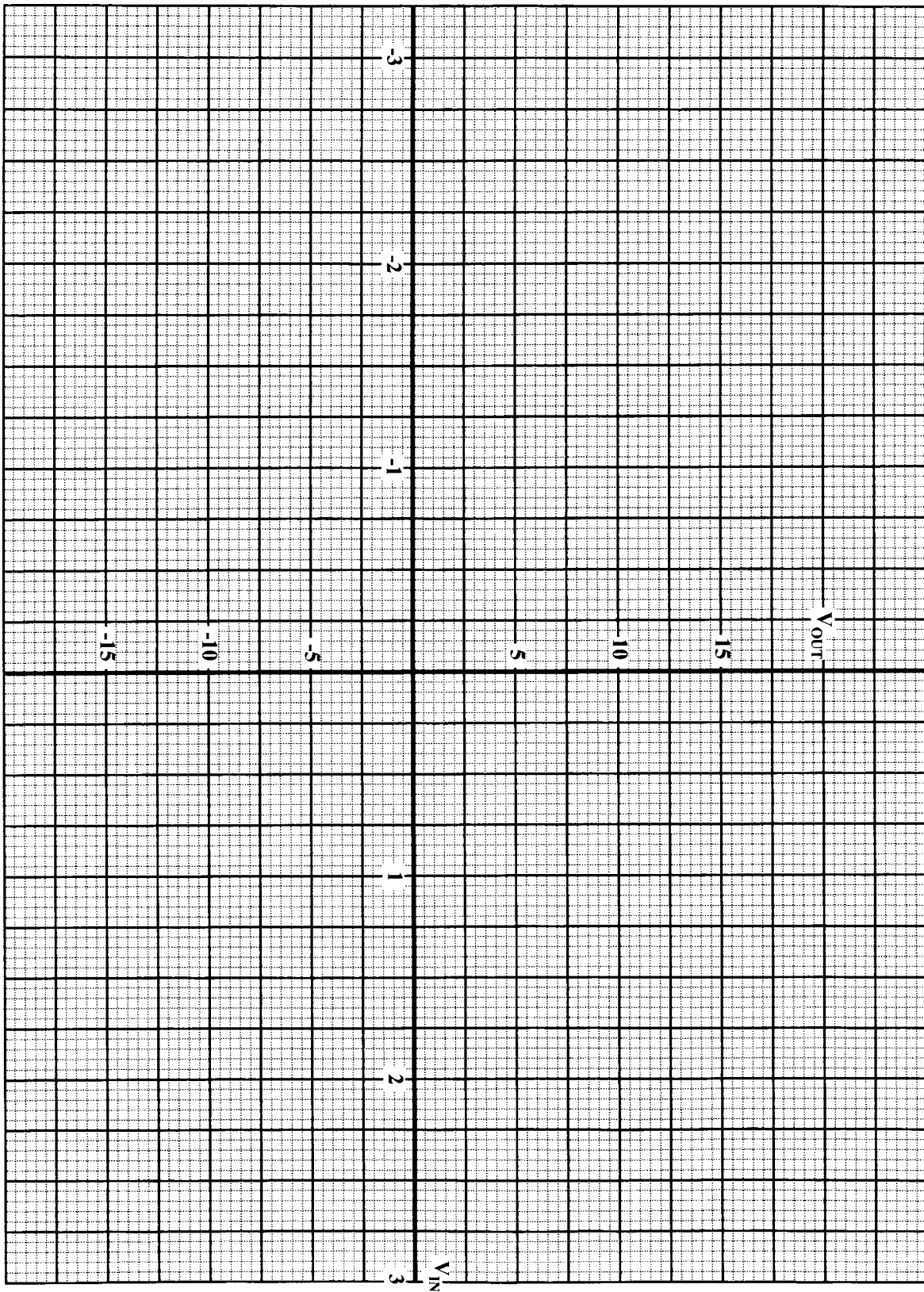
- (iii) What is the range of possible input voltages if the amplifier is NOT to be saturated?

Maximum positive input = _____

Maximum negative input = _____

[2 marks]

Total 15 marks



GO ON TO THE NEXT PAGE

3. (a) Figure 4 shows an experimental arrangement used to investigate the photoelectric effect of a metal.

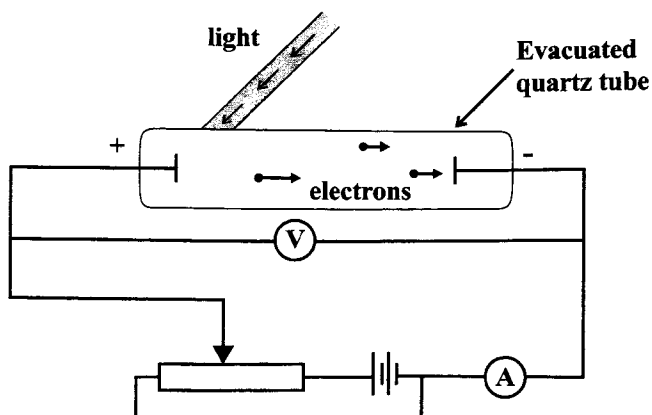


Figure 4. Experiment to investigate the photoelectric effect

The photoelectric equation may be written as $hf - \frac{1}{2}m_e v^2 = \phi$,
where h is Planck's constant, f is the frequency of the light used, ϕ is the work-function of the metal, m_e is the mass of the electron and v is the velocity of the emitted photoelectrons.

- (i) Explain what is meant by the 'stopping potential'.

[1 mark]

- (ii) With reference to Figure 4, explain how the stopping potential may be measured.

[2 marks]

- (b) When a certain metal is illuminated with light of different wavelengths, the stopping potentials in Table 2 are observed.

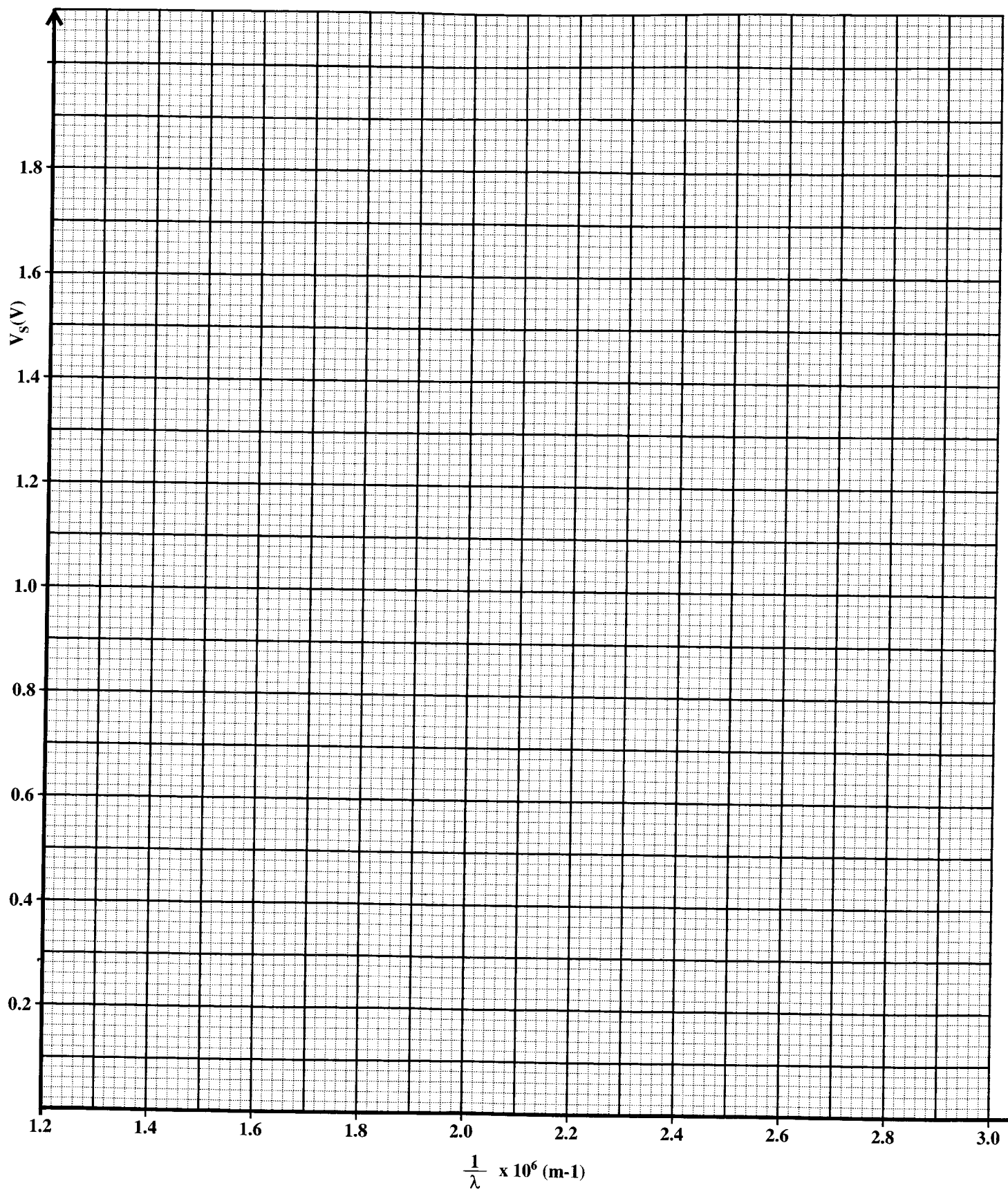
TABLE 2

Wavelength, λ (nm)	Stopping potential (V_s)	Reciprocal Wavelength ($1/\lambda$) (m^{-1})
366	1.48	
405	1.15	
436	0.93	
492	0.62	
546	0.36	
579	0.24	

- (i) Complete Table 2 by filling in the missing values of $\frac{1}{\lambda}$. [1 mark]
- (ii) On the grid on page 12, plot a graph of stopping potential, V_s against $\frac{1}{\lambda}$. [3 marks]
- (iii) Use the graph to determine
- a) Planck's constant

[4 marks]

GO ON TO THE NEXT PAGE



- b) the cut-off wavelength, λ_0

[2 marks]

- c) the work function of the metal.

[2 marks]

Total 15 marks

GO ON TO THE NEXT PAGE

SECTION B

Answer ALL questions.

You MUST write your answers in the answer booklet provided.

4. (a) The graph in Figure 5 shows the variation with temperature of the resistance of a thermistor.

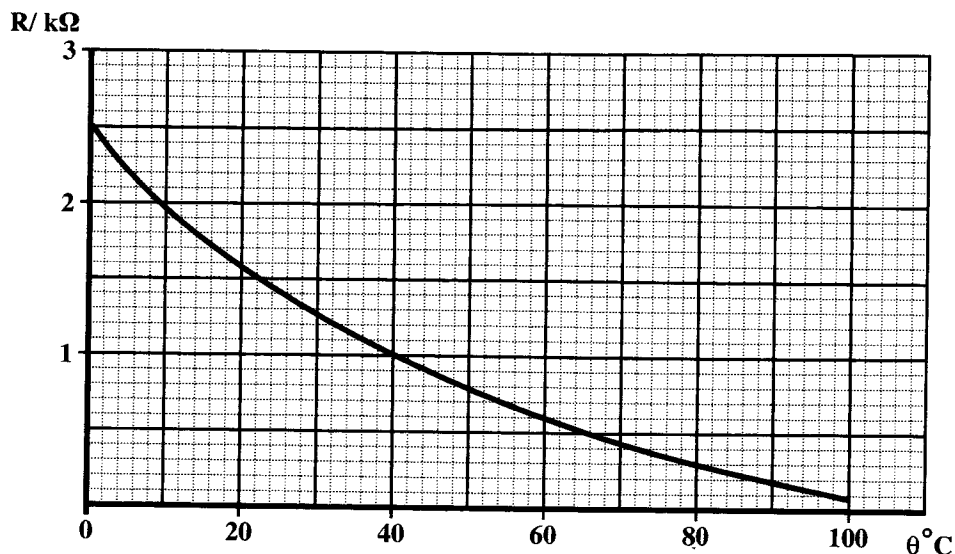


Figure 5. Variation of resistance with temperature for a thermistor

- (i) a) Describe how values for the graph in Figure 5 could be obtained.
- b) The diagram in Figure 6 shows a thermistor in a water bath. In your answer booklet copy Figure 6 and complete Figure 6 by drawing in the required circuit.

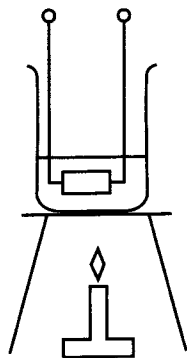


Figure 6

- (ii) The thermistor is included in a resistor network as shown (Figure 7).

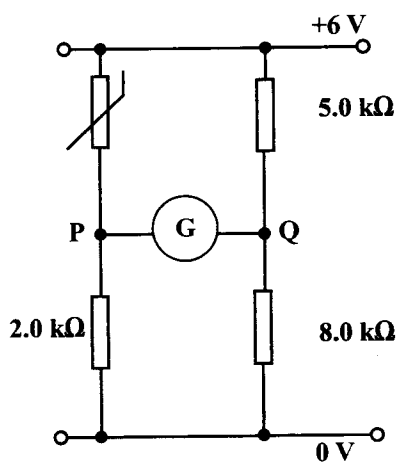


Figure 7

- a) Find the potential at point Q.
- b) At what thermistor temperature will the galvanometer read zero?
[11 marks]
- (b) In the circuit shown in Figure 8 the batteries have negligible internal resistance. The

currents supplied by the batteries are labelled x , y and z .

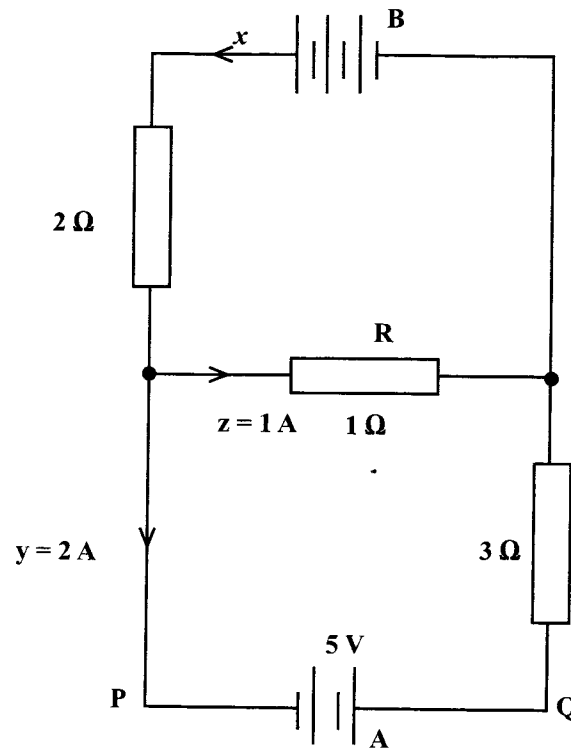


Figure 8

Given that current y is 2.0 A and current z is 1.0 A, calculate the e.m.f. of the battery labelled B.
[4 marks]

Total 15 marks

5. (a) (i) Draw the truth tables for a NOR gate and a NAND gate.

- (ii) Draw a circuit diagram to show how a NAND gate may be constructed from a number of NOR gates.
- (iii) Redesign the circuit shown in Figure 9 so that it may be constructed using ONLY NOR gates. Reduce the circuit to the MINIMUM chip count.

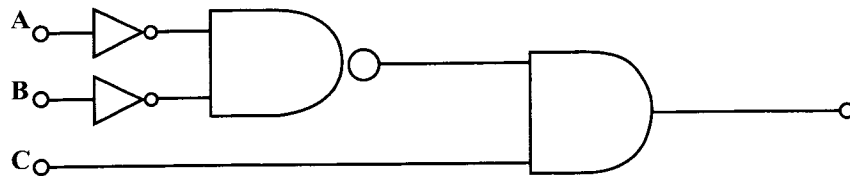


Figure 9

[8 marks]

- (b)
 - (i) Add the binary numbers 1011 and 101.
 - (ii) Discuss the difference between a half-adder and a full-adder in digital electronics.
 - (iii) Draw a diagram to show how a full adder is constructed from half-adders and give an example to explain its operation.

[7 marks]

Total 15 marks

6. (a) (i) What is the evidence in Millikan's oil drop experiment for the quantization of charge.
- (ii) Give TWO measures adopted by Millikan to improve accuracy in the oil drop experiment.
- (iii) Explain how it is possible to change the charge on an oil droplet.

[5 marks]

- (b) Figure 10 shows two plane parallel conducting plates 1.50×10^{-2} m apart, held horizontally, one above the other in air. The upper plate is maintained at a positive potential of 1.50×10^3 V while the lower plate is earthed. A small oil drop of mass 9.80×10^{-15} kg remains stationary in the air between the plates. The density of the air is negligible in comparison with that of oil.

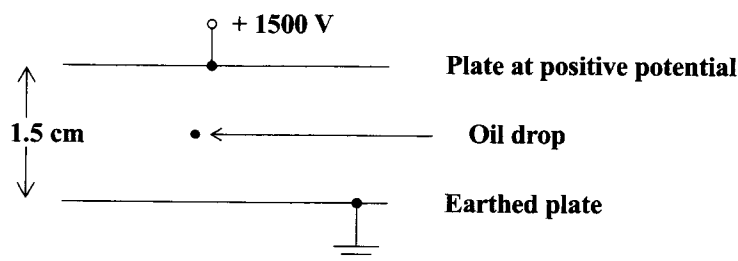


Figure 10

- (i) Draw a free body diagram showing the forces acting on the oil drop.
- (ii) Calculate
- a) the charge on the oil drop
- b) the number of electrons attached to the oil drop.
- (c) The potential of the upper plate is suddenly changed to -1.50×10^3 V. Determine the initial acceleration of the charged drop.

[10 marks]

Total 15 marks

END OF TEST